

1. Calibrating the Initial Economy

Goal: fit pre-2007 household moments. External inputs and all 12 targets and weights stay fixed.

1. **Try a parameter vector θ :** nine coefficients governing saving, fertility, housing, and bequests.
2. **Compute initial age-earnings shares** from survival, entry, and the earnings process. Set the pension to payroll revenue divided by retiree exposure; verify it against the solved distribution.
3. **Try a fertility preference ψ_0 and solve the housing equilibrium.** Guess the house price P , derive rent, solve household choices backward, and find the stationary distribution forward. Update P and repeat until housing demand equals supply.
4. **Update ψ_0 with a bracketed scalar search.** Repeat step 3 for each trial until model completed fertility equals 2.1 (within 0.0005).
5. **Score this parameter vector:** compute all 12 model moments and the weighted sum of squared model-target gaps.
6. **Update θ .** Perturb each coefficient to estimate how the moments respond; use these slopes for bounded, ridge-regularized joint steps. Redo steps 2–5 for each proposal, keep the lowest loss, and repeat within the search budget. Re-solve the selected winner twice.

Nested loops: parameter search $\longrightarrow$ fertility normalization $\longrightarrow$ housing-market clearing.

2. Fitting the Historical Transition

Hold the nine structural parameters fixed. Form 2007 age masses from data; within each age, keep the calibrated distribution of wealth, housing, and family states.

1. **Try ψ_{2007} , believed permanent.** Solve the terminal stationary economy at this preference level.
2. **Guess future price and pension paths.** Solve household choices backward from the terminal value; simulate household and person distributions and birth cohorts forward from the inherited 2007 state.
3. **Update the price and pension paths and repeat step 2** until housing clears and the payroll budget balances at every date. Verify the population accounting, recompute the path, and check a longer horizon.
4. **Change ψ_{2007} and repeat steps 1–3** until first-window fertility matches the 2008–2011 target of 1.974875 (error at most 0.005). Every trial starts from the same 2007 state.
5. **Advance only after acceptance.** Keep the first four-year outcome and birth queues. Repeat at 2011, 2015, and 2019 to target the following four years. Each new preference level is unanticipated.

After 2023, hold the final preference level fixed.

The household-rate analogue of female TFR is approximate; the complete history is not yet validated.